

E-Ink Info Dashboard — Cheat Sheet

Chain K — Hardware & Systems Foundations · one-page recall, not first-time learning

E-Ink Fundamentals

Term	Meaning
Bistable	Particles hold position with zero power once set — power only needed to CHANGE the image
Reflective	Uses ambient light, no backlight — sunlight-readable
Buying red flag	"e-paper" listing with no controller chip / driver library specified

Refresh & Ghosting

Mode	Behavior
Full refresh	Clears + redraws exactly, zero residual, several seconds
Partial refresh	Fast, region-only, moves each pixel only part-way per pass
Ghosting	Residual from partial refreshes that never fully converged
Fix for ghosting	Periodic full refresh — NOT more partial refreshes
Convergence math	$\text{error} \times (1 - \text{gain})$ per partial refresh, only against a FIXED target

Low-Power Service Design

Pattern	Rule
Scheduling	systemd timer + oneshot service — never a busy-loop or naive cron
Timer vs cron	Timer can catch up a missed run after boot; cron just skips it
Busy-loop waste	Polling faster than content/display can change wastes power + API quota

Layout Design

Rule	Detail
Region split	Static content (weather/calendar) = full refresh; dynamic (clock) = partial refresh
Panel size tradeoff	2.13" = clock/status only; 7.5" = full layout, slower full refresh
SPI control lines	MOSI, SCLK, CS, DC, RST, BUSY

Chain K · E-Ink-Info-Dashboard · full reasoning in LESSON.pdf